

Diabetes Classification and Clustering

1 Introduction

Diabetes is a major health condition associated with changes in blood glucose regulation and other metabolic factors. Machine learning can be used to analyse these measurements and identify patterns associated with different diabetes conditions.

The aim of this project is to build and compare classification and clustering models using a diabetes dataset with three classes:

- **N**: Non-diabetic
- **P**: Predict-diabetic
- **Y**: Diabetic

The assignment requires exploratory data analysis, model building, comparison of classification and clustering methods, interpretation of results, error analysis and possible future improvements.

2 Dataset Description

The dataset initially contained 1000 records and 14 columns. The features include demographic information and several metabolic measurements.

Table 1: Dataset Features

Feature	Description
Gender	Biological sex of the subject
AGE	Age in years
Urea	Blood urea level
Cr	Blood creatinine level
HbA1c	Glycated hemoglobin level
Chol	Total cholesterol
TG	Triglycerides
HDL	High-density lipoprotein
LDL	Low-density lipoprotein
VLDL	Very-low-density lipoprotein
BMI	Body mass index
CLASS	Diabetes class (N, P or Y)

The columns `ID` and `No_Pation` were identifiers and were not used as predictive features.

3 Methodology

The overall workflow used in the project was:

Data Loading → Data Cleaning → EDA → Preprocessing → Classification →
Clustering → Evaluation → Error Analysis

3.1 Data Cleaning

The dataset was first inspected for its structure, data types, unique values and basic statistics. The identifier columns `ID` and `No_Pation` were removed.

The `Gender` column contained both upper- and lower-case values. These values were standardised and encoded numerically, with Male represented by 0 and Female by 1. Extra spaces in the `CLASS` column were also removed.

Duplicate records were checked before modelling. A total of 174 duplicate rows were identified and removed, reducing the dataset from 1000 to 826 records.

The numerical variables were then checked for values less than or equal to zero. One invalid value was found in the `Chol` feature, where cholesterol was recorded as 0. This record was removed. The final dataset contained 825 records.

3.2 Class Distribution

The final class distribution was highly imbalanced:

Table 2: Class Distribution

Class	Percentage	Description
Y	83.64%	Diabetic
N	11.52%	Non-diabetic
P	4.85%	Predict-diabetic

The large difference between the class sizes was considered during model training because a model could otherwise perform well on the majority class while performing poorly on the smaller classes.

3.3 Exploratory Data Analysis

EDA was carried out before model training to understand the distribution of the variables and differences between classes.

Boxplots were created for the numerical features against the three diabetes classes. Histograms were used to inspect the individual feature distributions. A correlation matrix was also generated to examine relationships between the numerical variables.

The EDA showed that the dataset contains considerable variation in several measurements, particularly Urea and creatinine. Some features also contain relatively large values compared with their typical ranges, which can affect distance-based methods and therefore made feature scaling important.

4 Data Preprocessing

The CLASS column was separated from the remaining variables to form the target variable and input features.

The data was split into training and testing sets using an 80:20 ratio. Stratification was used so that the class proportions were maintained between the two sets. The test set contained 165 records.

For models that are sensitive to feature scale, StandardScaler was applied. The scaler was fitted on the training data and then used to transform the training and test sets.

The class labels were converted into numerical values using LabelEncoder. Class weights were also used during classification to reduce the effect of the class imbalance.

5 Classification Models

Four supervised learning models were trained:

1. Logistic Regression
2. Random Forest
3. Support Vector Machine (SVM)
4. Artificial Neural Network (ANN)

5.1 Logistic Regression

Logistic Regression was used as a simple baseline model. It provides a linear decision boundary and gives a useful reference for comparing the more complex models.

The model was trained with balanced class weights and a maximum iteration limit of 2000.

5.2 Random Forest

Random Forest was used as the main tree-based classification model. It was configured with 300 decision trees and balanced class weights.

Random Forest was also used for feature importance analysis, allowing the contribution of individual features to be examined.

5.3 Support Vector Machine

A Support Vector Machine was trained using the scaled features. Class weights were applied to account for the imbalance between N, P and Y.

5.4 Artificial Neural Network

A feed-forward Artificial Neural Network was implemented using three hidden layers. The hidden layers contained 32, 24 and 16 neurons respectively, using ReLU activation.

Dropout was included between the layers to reduce overfitting. The output layer used Softmax activation to produce probabilities for the three classes.

The network was trained using the Adam optimizer and sparse categorical cross-entropy loss. Training was limited to 100 epochs, with early stopping based on validation loss. Class weights were also supplied during training.

6 Classification Evaluation

The models were evaluated using accuracy, precision, recall and F1-score. Macro-average precision, recall and F1-score were used for comparison because the class distribution was highly imbalanced. Confusion matrices were also generated to inspect individual class predictions.

Table 3: Classification Model Performance

Model	Accuracy	Precision	Recall	F1
Random Forest	98.18%	0.993	0.899	0.940
ANN	93.33%	0.773	0.834	0.798
SVM	92.73%	0.774	0.832	0.791
Logistic Regression	91.52%	0.770	0.842	0.788

Random Forest produced the best overall accuracy at 98.18%. Its macro F1-score was also the highest among the tested models.

The detailed performance of the Random Forest model was:

Table 4: Random Forest Classification Report

Class	Precision	Recall	F1-score	Support
N	1.000	0.947	0.973	19
P	1.000	0.750	0.857	8
Y	0.979	1.000	0.989	138
Macro Average	0.993	0.899	0.940	165
Weighted Average	0.982	0.982	0.981	165

The model classified all tested P cases that it predicted as P correctly, giving P a precision of 1.00. However, its recall for P was 0.75, meaning some actual P cases were classified as Y.

7 Feature Importance

Feature importance from the Random Forest model was used to identify the variables that contributed most to its predictions.

Table 5: Random Forest Feature Importance

Feature	Importance
HbA1c	0.4619
BMI	0.1883
AGE	0.1419
Chol	0.0438
TG	0.0374
Urea	0.0296
VLDL	0.0251
Cr	0.0235
LDL	0.0205
HDL	0.0176
Gender	0.0103

HbA1c had the highest feature importance by a large margin, followed by BMI and AGE. This indicates that these features were particularly useful for separating the diabetes classes in the Random Forest model.

8 Clustering

Unlike classification, clustering was performed without using the CLASS labels during the clustering process. The objective was to determine whether the natural structure of the feature data could separate the three diabetes groups.

Before clustering, all features were standardised. Principal Component Analysis (PCA) was then applied to reduce the dimensionality of the data.

8.1 PCA

The number of principal components was selected so that at least 90% of the variance was retained. Nine components were required.

The original feature matrix had 11 features, while the PCA-transformed dataset contained 9 components. These components retained 93.08% of the total variance.

The PCA-transformed data was then used for clustering.

8.2 Clustering Algorithms

Three clustering algorithms were tested:

- K-Means
- Agglomerative Clustering
- HDBSCAN

K-Means and Agglomerative Clustering were configured to produce three clusters. HDBSCAN was allowed to determine the cluster structure using a minimum cluster size of 15.

The resulting clusters were compared with the known class labels using Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI).

9 Clustering Evaluation

Table 6: Clustering Performance

Algorithm	ARI	NMI
K-Means	0.204	0.305
Agglomerative Clustering	0.537	0.348
HDBSCAN	0.004	0.012

Agglomerative Clustering produced the best clustering result, with an ARI of 0.537 and NMI of 0.348. K-Means showed a weaker relationship with the actual classes, while HDBSCAN produced very low scores.

The clustering results were considerably weaker than the classification results. This suggests that the three diabetes classes are not separated into three clearly distinct groups based only on the available feature space.

10 Classification vs. Clustering

The two approaches produced noticeably different results.

Classification had access to the known class labels during training, allowing the models to learn the relationship between the medical features and the target classes. Random Forest achieved 98.18% accuracy.

Clustering did not use the class labels. Instead, it attempted to discover groups from the feature values alone. The best clustering method, Agglomerative Clustering, achieved an ARI of 0.537.

Therefore, classification was much more effective for this dataset. The difference is also expected because the “P” class represents a transitional or predictive category, which may not form a completely separate natural cluster from the N and Y groups.

11 Error Analysis

Error analysis was performed using the best classification model, Random Forest.

There were two misclassified cases involving the P class. Both cases were actual P cases but were predicted as Y.

Table 7: Misclassified P Cases

HbA1c	BMI	Actual	Predicted
6.3	27.0	P	Y
5.8	32.0	P	Y

These cases show the difficulty of separating the Predict-diabetic class from the Diabetic class. Their feature values can overlap with the characteristics of the other classes.

The confusion matrices also showed that the majority class Y was classified very well. However, because P contains far fewer samples, even a small number of errors has a noticeable effect on its recall and F1-score.

12 Discussion

The results show that Random Forest was the strongest model among the tested classification approaches. Its high performance can partly be attributed to its ability to model non-linear relationships between the different medical measurements.

The ANN, SVM and Logistic Regression models also performed reasonably well, but their overall accuracies were lower. The difference was particularly visible for the smaller classes.

The class imbalance remained an important challenge. Although class weighting was used, the P class still had lower recall than the Y class. This is important because the P class is the transition category and may be more difficult to distinguish from the other two groups.

For clustering, Agglomerative Clustering performed better than K-Means and HDBSCAN. However, its ARI of 0.537 indicates only moderate agreement with the known class structure. The low HDBSCAN score suggests that its density-based assumptions did not match the structure of this dataset well.

13 Limitations

Several limitations should be considered when interpreting the results:

- The dataset is strongly imbalanced, with the Y class making up more than 80% of the final data.
- The P class contains relatively few observations, making it harder to learn its characteristics reliably.
- Some medical measurements contain extreme values, which may influence both classification and clustering.
- The dataset is relatively small after duplicate and invalid records are removed.
- The clustering methods were evaluated against known labels, but the actual clustering process itself did not use those labels.
- A single train-test split was used for the classification experiments, so the reported results may vary with a different split.

14 Future Enhancements

The project could be improved in several ways.

First, cross-validation could be used instead of relying on a single train-test split. This would provide a more stable estimate of model performance.

The class imbalance could also be investigated further using techniques such as over-sampling or SMOTE, while carefully avoiding data leakage.

Hyperparameter tuning could be performed for Random Forest, SVM and the neural network. Different tree counts, tree depths, SVM kernels and neural network architectures could be tested.

For clustering, additional methods such as Gaussian Mixture Models, DBSCAN with tuned parameters, or other dimensionality reduction techniques could be explored. The clustering features could also be selected based on their relevance instead of using all available variables.

Finally, additional medical or demographic features and a larger dataset could improve the ability to distinguish the Predict-diabetic class from the other two groups.

15 Conclusion

This project compared supervised classification and unsupervised clustering for a three-class diabetes dataset.

The data was cleaned by removing identifier columns, duplicate records and invalid rows. EDA was then performed to understand feature distributions, class imbalance and correlations. The processed data was used to train four classification models and three clustering algorithms.

Among the classification models, Random Forest performed best with an accuracy of 98.18% and a macro F1-score of 0.940. HbA1c was the most important feature according to the Random Forest feature importance analysis, followed by BMI and AGE.

For clustering, Agglomerative Clustering performed best with an ARI of 0.537 and NMI of 0.348. However, the clustering results were substantially weaker than the classification results.

Overall, the results indicate that the available features contain enough information for strong supervised classification, but the natural structure of the data does not separate the three diabetes classes as clearly when the labels are not provided.